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Revolutionary H₂S Scavenging Technology Transforming Safety, Sustainability, and Performance in Gas Processing

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Abstract

This presentation outlines an innovative H₂S scavenger program which has demonstrated exceptional performance in operational safety enhancement and carbon footprint reduction. The program is aimed to controlling H₂S levels across multiphase and gas export lines to achieve target KPIs of <10ppm H₂S in gas phase

Daleel Petroleum with the support of Veolia technology Experts of Veolia utilized specialized chemistries developed to generate maximum scavenging capacity. The product selection process commenced with comprehensive laboratory evaluations, incorporating extensive testing under simulated field conditions and detailed compatibility. Following successful laboratory validation, Veolia's proprietary modelling was used to estimate the required chemical volume to achieve target KPI.

Strategic deployment of the non-amine Scavenger technology was implemented at key selected locations to maximize scavenging efficiency. The field implementation phase utilized performance monitoring tools for continuous evaluation.

ProSolv HS8796 scavenger chemistry successfully reduced H₂S concentrations in multiphase streams from 6-8mol % to parts per million levels and eventually to below 5ppm at gas export lines achieving compliance with sales gas specifications and contractual obligations. The performance of the product shown much higher scavenging capacity over previously applied scavenger chemistries.

This methodical approach has yielded multiple benefits including superior H₂S scavenging performance, enhanced operational safety for personnel and surrounding communities, and significant environmental improvements through supporting the elimination of off-spec gas flaring. Additionally applying the novel scavenger chemistry enabled a reduction of approximately 23K tons/yr. of CO₂ which is translated into approximately 7K medium size cars removed from the road.

Background

Sour oil production represents a significant challenge in the petroleum industry [1], characterized by the presence of high concentrations of hydrogen sulfide (H₂S) and carbon dioxide (CO₂). This phenomenon has become increasingly prevalent as mature fields deplete and new discoveries often contain higher levels

of acidic compounds. The presence of these compounds poses multiple operational challenges, including severe corrosion of equipment and infrastructure, health and safety risks to personnel, environmental concerns, and reduced product quality.

The corrosive nature of sour hydrocarbons can lead to premature failure of well tubulars, processing equipment, and pipeline systems, resulting in increased maintenance costs and potential production disruptions [2]. Additionally, the presence of H₂S presents significant safety hazards, requiring sophisticated monitoring systems and emergency response protocols. Environmental regulations regarding sulfur emissions have also become more stringent, necessitating complex treatment processes.

Mitigation strategies typically involve a multi-faceted approach, including materials selection (such as corrosion-resistant alloys), chemical treatment programs (including H₂S scavengers and corrosion inhibitors), and process modifications (such as gas sweetening units).

Advanced monitoring techniques, including real-time corrosion monitoring and predictive maintenance programs, have been implemented to manage asset integrity. The economic impact of sour oil production extends beyond direct operational costs to include specialized equipment requirements, enhanced safety measures, and more complex processing facilities. Understanding and effectively managing these challenges is crucial for maintaining safe and economically viable operations in sour oil fields.

Operational Challenges and Mitigation Strategies

Daleel Petroleum, a prominent and rapidly expanding energy producer in the Middle East, is currently experiencing significant operational expansion through the development of sour hydrocarbon reservoirs characterized by elevated hydrogen sulfide (H₂S) concentrations, reaching critical levels of 6-8mol%. This strategic expansion into sour reservoirs presents substantial technical and operational challenges that require comprehensive management strategies.

The company's operational portfolio now encompasses reservoirs with increasingly complex chemical compositions, particularly elevated H₂S concentrations. These heightened levels present multifaceted challenges across various operational domains, necessitating robust safety protocols and sophisticated processing solutions.

Daleel Petroleum has implemented a comprehensive sour stream management strategy encompassing multiple technical domains. The strategy incorporates proactive reservoir souring management protocols, enhanced materials of construction and upgrades, advanced sour gas stripping technologies, implementation of fixed-bed treatment systems, and strategic deployment of chemical scavenging solutions [3,4].

The company's established treatment protocol has historically centered on non-amine H₂S scavenger chemistry, strategically deployed across multiphase process streams. The treatment cascade comprises primary phase multiphase stream treatment and secondary phase post-separation processing, which includes sorbent-based fixed bed treatments, amine-type polishing H₂S scavenger applications, and strategic stream dilution with sweet streams [5].

Operations are under stringent contractual obligations requiring H₂S concentrations not exceeding 5 ppm at the sales gas export point. This compliance requirement drives the company's aggressive H₂S reduction strategy at the multiphase inlet stage of the gas processing facility.

Failure to maintain H₂S levels below the specified threshold presents several significant risks. From a Health, Safety, and Environmental (HSE) perspective, concerns include personnel safety within operational facilities, potential community health impacts in surrounding areas, and environmental compliance challenges. Technical and operational impacts encompass formation of iron sulfide deposits throughout the produced water reinjection network, potential reservoir damage and reduced injectivity, and increased maintenance requirements and system reliability concerns.

Commercial and regulatory consequences include financial penalties for non-compliant gas exports, potential production constraints and operational limitations, and contractual compliance challenges.

Environmental sustainability considerations extend to increased carbon footprint due to necessary flaring of off-specification gas, environmental regulatory compliance challenges, and impact on corporate sustainability metrics.

This operational scenario underscores the critical importance of maintaining effective H₂S management strategies while balancing operational efficiency, safety requirements, and environmental responsibilities. Daleel Petroleum's approach demonstrates the complex interplay between technical solutions, operational requirements, and regulatory compliance in modern hydrocarbon production operations.

Site-Specific Chemical Evaluation and Field Validation

Field Survey and Chemistry Evaluations involve a comprehensive assessment conducted jointly by Veolia and Daleel Petroleum teams to analyze process operations and production characteristics. This evaluation encompasses multiple critical aspects of the operation.

The on-site data collection phase focuses on gathering essential operational parameters including operating pressure and temperature measurements, flow rates and production volumes, current H₂S levels and variations, and the effectiveness of existing treatment methods. Representative operating conditions are in [table 1](#) below

Table 1—Typical Field operating conditions

Item	Value
Oil Density	API 24-37
Temperature	15-50 C
Pressure	100-260 Psi
Water Cut	70-85%
H₂S levels (measured at the gas phase in contact with liquids)	> 40,000 ppm
Produced Water pH	6.0-6.95

During the process analysis phase, teams conduct thorough reviews of current scavenging systems, identify operational bottlenecks, assess equipment conditions, and evaluate any process limitations that may affect performance.

The chemical assessment component involves analyzing the performance of current H₂S scavengers, evaluating potential alternative non-amine scavenger options, conducting compatibility studies with existing infrastructure, and determining chemical reaction efficiencies under various conditions.

Operational requirements are carefully considered through determination of optimal dosing rates, assessment of storage and handling needs, evaluation of safety considerations for new chemical deployment, and comprehensive cost-benefit analysis of proposed changes.

Environmental considerations form a crucial part of the evaluation, encompassing environmental impact assessments, waste management requirements, and ensuring regulatory compliance across all proposed modifications.

This comprehensive evaluation serves as the foundation for determining the feasibility and requirements for implementing a higher capacity H₂S scavenging solution while maintaining operational efficiency and safety standards.

Based on site provisions and available injection points, a specialty Veolia H₂S scavenger ProSolv HS8796 was proposed for evaluation against incumbent H₂S scavenger for comparative performance, compatibility with the system fluids, material of construction and potential solid formation.

ProSolv HS8796 is a branched alcohol-based chemistry that when compared to commercially available non-amine-based chemistries, has a very high capacity for scavenging H₂S. ProSolv HS8796 works by chemically reacting with hydrogen sulfide (H₂S) to form stable, non-toxic compounds. It is effective at removing H₂S from gas streams and liquid hydrocarbons through a neutralization reaction. The specialized chemistry offers advantages like relatively fast reaction kinetics, high efficiency at low concentrations, and good thermal stability. ProSolv HS8796 is successfully applied by Veolia customers globally at various geographies.

Laboratory Analysis and Performance Validation Results

H₂S Removal Efficiency: Performance Analysis and Validation

Veolia conducted high-temperature, high-pressure vessel evaluations comparing ProSolv HS8796 against the incumbent non-amine based H₂S scavenger. Test conditions simulated field parameters: 70% water cut, 90°C, and 150 psig.

Daleel's expansion activities and increased sour stream production require higher-capacity scavenging solutions while utilizing existing injection infrastructure. Implementation of ProSolv HS8796 offers potential optimization of logistics and reduced chemical mobilization frequency.

Laboratory results confirm ProSolv HS8796's enhanced scavenging capacity, supporting progression to field deployment and optimization initiatives. These findings align with operational objectives for improved H₂S management efficiency.

Figure 1 demonstrates ProSolv HS8796's superior scavenging capacity under simulated field conditions. The enhanced performance characteristics provide quantitative validation for field application.

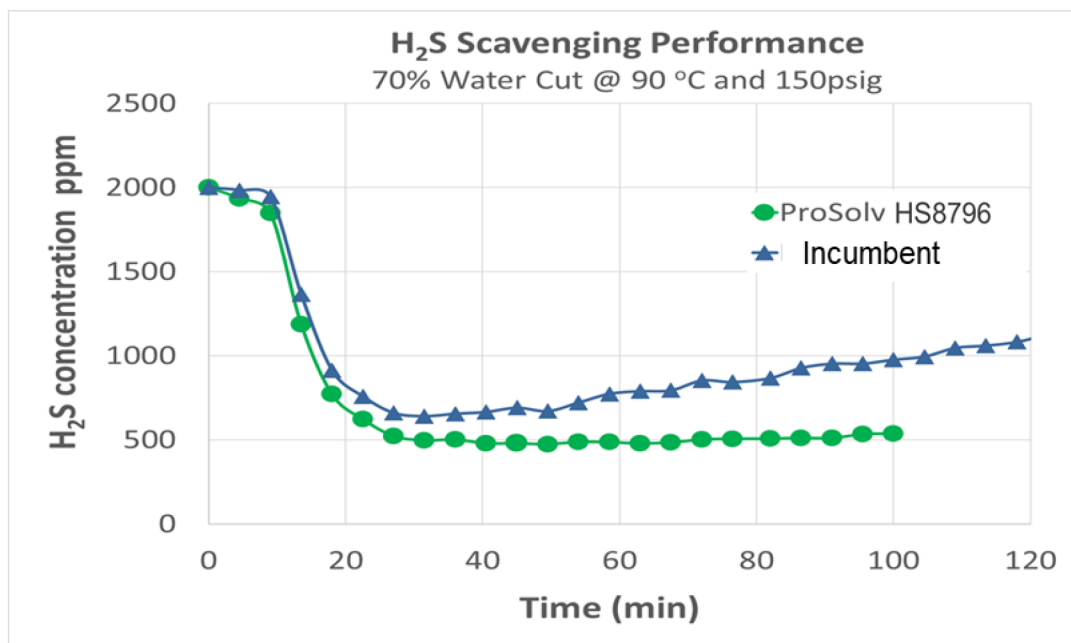


Figure 1—Comparison between H₂S scavenger capacity between ProSolv HS8796 and incumbent H₂S scavenger.

Materials of Construction: Compatibility Validation Study

A comprehensive materials compatibility evaluation was conducted for ProSolv HS8796 to assess its interaction with existing infrastructure materials in Daleel's operations. This assessment was crucial to ensure long-term operational integrity and equipment reliability [6].

The compatibility study encompassed systematic testing of ProSolv HS8796 with various materials of construction commonly present in Daleel's processing facilities. The evaluation protocol included extended exposure testing, corrosion rate measurements, surface interaction analysis, material degradation assessment, and stress corrosion susceptibility evaluation.

As presented in Table 2, the compatibility evaluation yielded detailed results across multiple material categories. The assessment considered carbon steel components, stainless steel grades, elastomeric materials, non-metallic components, and specialty alloys. The results demonstrate material-specific responses under various operating conditions, including normal operating temperatures, maximum upset conditions, various concentration levels, and different exposure durations.

Table 2—Material compatibility with ProSolv HS8796.

Material	Test Results
304 Stainless Steel	Compatible
316 Stainless Steel	Compatible
Low Carbon Steel	Compatible
HDPE	Compatible

The compatibility data provides critical insights for equipment lifetime predictions, maintenance scheduling optimization, material selection validation, risk assessment protocols, and operational safety parameters.

The compatibility evaluation confirms ProSolv HS8796's suitability for deployment within Daleel's existing infrastructure, supporting implementation decisions while ensuring operational integrity and safety standards are maintained. The results establish a foundation for confident integration of ProSolv HS8796 into existing processes while maintaining system reliability and operational excellence.

Chemical Compatibility Analysis: ProSolv HS8796 and Incumbent Scavenger Integration

In accordance with the Management of Change (MOC) protocol, Veolia performed a comprehensive compatibility assessment between ProSolv HS8796 and the existing H2S scavenger. This crucial evaluation aimed to verify seamless integration capabilities within current chemical management systems.

The lab evaluation methodology encompassed a detailed examination of direct chemical interactions under strictly controlled laboratory conditions. The testing procedure specifically involved combining the pure products in equal proportions to accurately replicate potential field mixing scenarios.

The results, as evidenced in Figure 2, conclusively demonstrated optimal compatibility characteristics. The evaluation confirmed complete miscibility and stability between ProSolv HS8796 and the incumbent scavenger, with no evidence of adverse chemical reactions, phase separation, or precipitate formation upon mixture. These findings strongly support the successful implementation of ProSolv HS8796 within existing operational frameworks.



Figure 2—Compatibility evaluations between incumbent and ProSolv HS8796 H₂S scavengers.

The demonstrated compatibility results carry significant operational implications, particularly concerning the field implementation of ProSolv HS8796 in site-specific H₂S scavenger day tanks. The validated compatibility ensures optimal operational flexibility throughout the transition phase while substantially reducing potential risks associated with chemical handling and injection procedures.

Furthermore, this comprehensive compatibility assessment conclusively establishes the viability of incorporating ProSolv HS8796 into current chemical management systems without disrupting ongoing operations. The findings provide strong evidence supporting the safe and efficient implementation of this new chemical solution while maintaining the integrity of existing systems and ensuring continued operational effectiveness. This validation represents a crucial milestone in the successful integration of ProSolv HS8796 into existing industrial processes.

Production Fluid Compatibility Assessment and Analysis

The team conducted a thorough compatibility analysis of ProSolv HS8796 with field-produced water across multiple dosage concentrations. The results of these evaluations, documented in [Figure 3](#), demonstrated conclusive evidence of complete compatibility between ProSolv HS8796 and the production stream water samples collected from various treated streams.



Figure 3—Compatibility evaluations between ProSolv HS8796 and production stream at various dose rates.

This comprehensive assessment, performed across different dose rates, confirmed that ProSolv HS8796 integrates seamlessly with produced water without any formation of solids or adverse reactions. The compatibility validation is particularly crucial as it ensures the product's safe and effective deployment in field operations, eliminating potential risks of operational disruptions or complications during actual implementation.

pH Stability Analysis of Treated Production Fluids

Additional evaluation was conducted to assess the potential pH impact of ProSolv HS8796 on produced water samples across various dosage concentrations.

The analysis, as documented in Table 3, revealed that the introduction of ProSolv HS8796 did not significantly alter the pH levels of the treated produced water under the tested conditions.

Table 3—pH of produced water versus dose rate of ProSolv HS8796.

	Dose rate of ProSolv HS8796 (ppm)			
Sample	Blank	500	1000	2000
PH Value	6.54	6.42	6.37	6.33

This stability in pH values across different dose rates demonstrates the chemical's neutral impact on the water chemistry, which is a crucial factor for maintaining process integrity and preventing potential corrosion or scaling issues in production systems. These findings further validate ProSolv HS8796's suitability for field application, as it maintains the natural chemical balance of the produced water while performing its intended function.

As per pH values, the slight PH impact of the product is helping to shift the balance toward more H₂S (instead of HS⁻) before gas-liquid separation. This has shown to help reduce how much HS⁻ ends up in the water phase, which is known to react strongly with iron and contribute to FeS formation during water treatment.

Solid formation Risk Assessment

A critical evaluation was conducted to assess ProSolv HS8796's potential for solid formation, a significant flow assurance concern commonly associated with H₂S scavengers, particularly those containing amine-based chemistries.

To thoroughly investigate this risk, lab evaluations performed extensive testing using produced water samples treated with ProSolv HS8796 at various concentrations, including extreme dosage levels up to 8000 ppm. The testing protocol involved careful visual monitoring of the treated samples, specifically focusing on detecting any signs of solid precipitation or sediment accumulation at the sample containers' bottom.

This assessment was essential to ensure that ProSolv HS8796 would not contribute to flow assurance issues that could potentially compromise operational efficiency and system integrity. The rigorous testing at elevated concentrations demonstrates the commitment to thorough risk assessment and operational safety.

Laboratory analysis results, as illustrated in [Figure 4](#), conclusively demonstrate that ProSolv HS8796 exhibits no tendency to form solid precipitates when mixed with produced water samples, even when administered at exceptionally high concentrations of up to 8000 ppm.

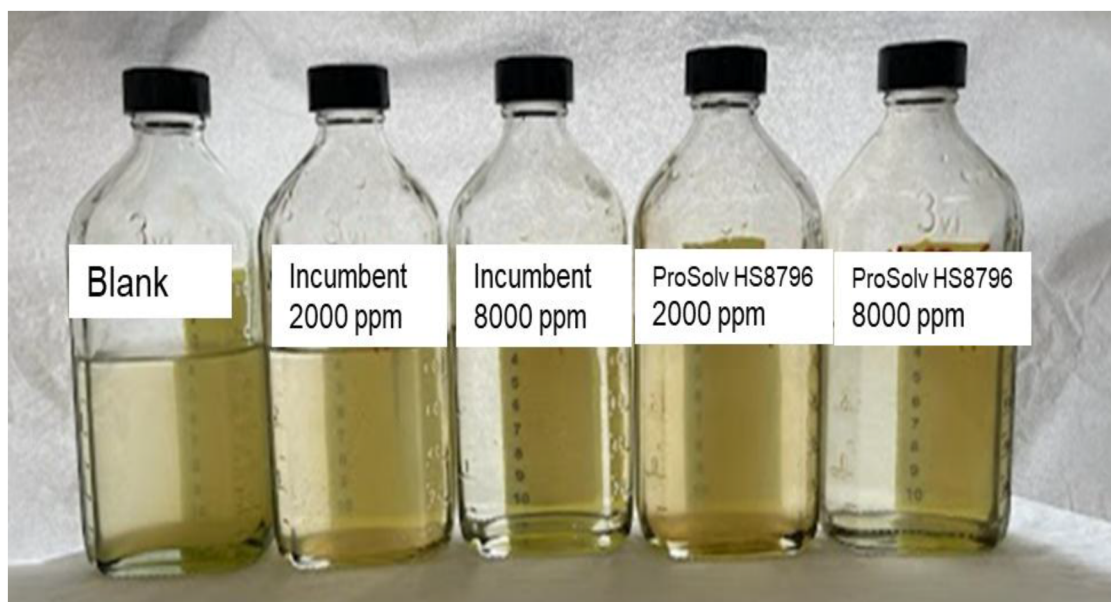


Figure 4—Solid formation evaluations

This significant finding confirms the product's superior stability characteristics and its ability to maintain solution clarity without generating problematic solid formations. Such performance characteristics represent a notable advantage over conventional H₂S scavengers, particularly in terms of flow assurance and operational reliability.

The absence of solid formation even at extreme dosage levels provides strong evidence for ProSolv HS8796's safe and efficient application in field operations without risking system blockages or flow restrictions.

Field implementation results and performance validation

Following the various lab evaluations, Veolia and Daleel determined to implement ProSolv HS8796 utilizing injection rates identical to those of the existing scavenger product. This strategic approach was designed to enable a direct comparison of ProSolv HS8796's H₂S scavenging effectiveness under equivalent treatment conditions.

The implementation plan includes a future phase for dose optimization once the application demonstrates stable performance. The operational setup and monitoring protocol are detailed in [Figure 5](#)'s schematic

flow diagram, which illustrates both the scavenger injection points and the designated H₂S sampling locations. This systematic approach ensures accurate performance assessment while maintaining operational consistency during the transition phase.

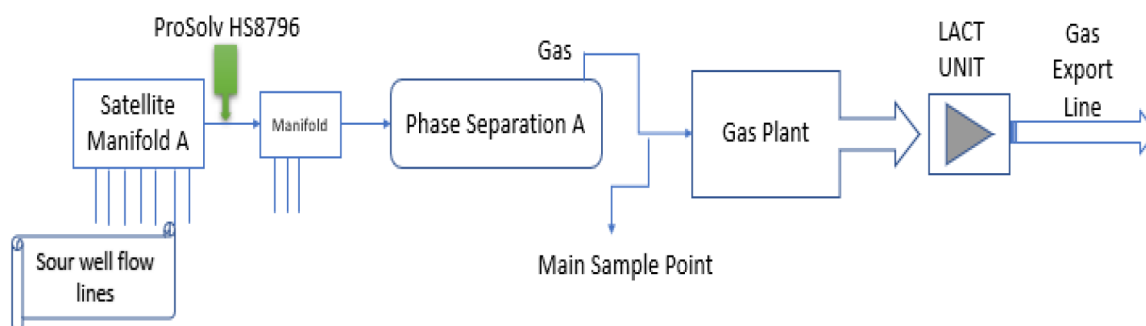


Figure 5—schematic for process flow, chemical injection and H₂S level sample point.

During the implementation of ProSolv HS8796, a rigorous monitoring protocol was established utilizing stain tubes to measure H₂S concentrations in the vapor phase of liquid samples, following ASTM D5705 standards [7].

Chemical dosage volumes underwent regular optimization to maintain H₂S levels at approximately 150 ppm at the gas plant inlet. The process benefits from additional downstream treatment within the gas plant facility, which further reduces H₂S concentrations to meet the critical Key Performance Indicator (KPI) requirement of less than 5 ppm in the gas export line. This systematic approach to measurement and optimization ensures consistent performance monitoring while maintaining compliance with operational specifications throughout the treatment process [8].

Performance data presented in Figure 6 provides compelling evidence of ProSolv HS8796's superior efficacy compared to the incumbent product.

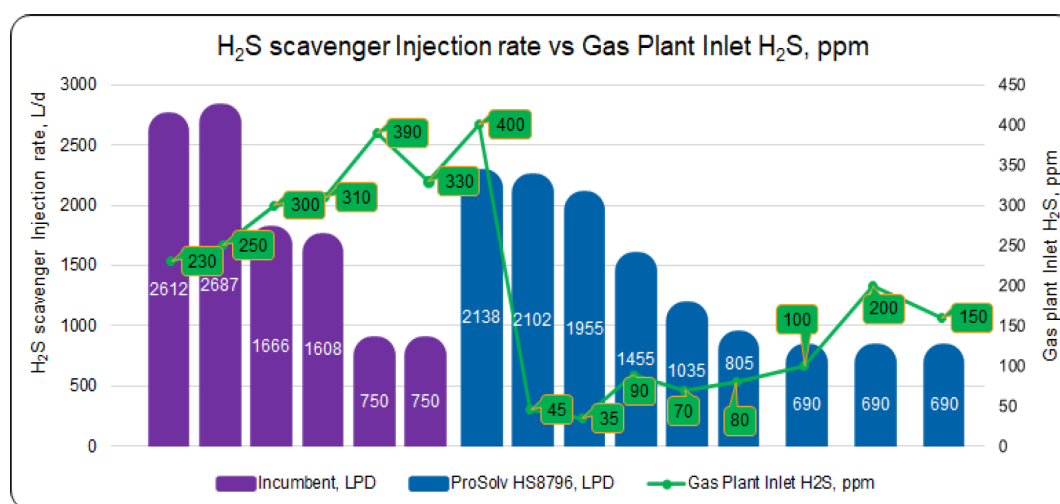


Figure 6—H₂S levels at gas plant inlet versus scavenger injection rates.

When administered at comparable injection rates, ProSolv HS8796 demonstrated significantly enhanced H₂S reduction capabilities at the gas plant inlet. Most notably, the scavenging efficiency of ProSolv HS8796 proved to be approximately two to three times more effective than the incumbent product while maintaining the same injection volumes.

This remarkable improvement in performance efficiency clearly establishes ProSolv HS8796 as a more potent H₂S scavenging solution, offering substantial advantages in treatment effectiveness while utilizing similar chemical volumes.

Environmental Performance Metrics and GHG Reduction Analysis

The implementation of ProSolv HS8796 has resulted in significant operational and environmental improvements across multiple aspects of the treatment process. The system achieved required KPIs while delivering several key advantages: reduced chemical consumption compared to the incumbent product, elimination of downstream triazine-based chemistry applications, and decreased transportation requirements for chemical delivery. Additionally, improved pH stability and H₂S control have reduced iron sulfide (FeS) formation and other sedimentation problems, which often lead to frequent pigging, chemical flushing, and process downtime. By reducing these interventions, the system supports more sustainable operations and contributes to lower GHG emissions. EPA Estimates that the average passenger vehicle emission rate of 4.6 metric tons of CO₂ per year [9].

Using Veolia's proprietary database for environmental impact assessment, a comprehensive analysis of greenhouse gas (GHG) emissions was conducted, considering factors such as scavenger chemistry, packaging materials, manufacturing processes, and transportation [10].

The comparative analysis, detailed in [Tables 4 and 5](#), presents daily equivalent CO₂ emissions for both the historical treatment method and the current ProSolv HS8796 system.

Table 4—Daily Overall GHG emissions using historical and incumbent treatment

Historical Case	Value	Unit	Emission Factor	Emission Factor Unit	GHG emission (kgCO ₂ e)
Incumbent H ₂ S scavenger	3416	kg	1.34	kgCO ₂ e/kg	4,585.84
Totes (IBC) 1000L	1058	Unit	126.7	kgCO ₂ e/Unit	134,05
Triazine based H ₂ S Scavenger	9467	kg	0.912	kgCO ₂ e/kg	863.12
Transport, freight, lorry, road trips	36090	metric ton*km	0.14	kgCO ₂ e/metric ton*km	4,930.17
Daily overall GHG Emissions					144,43

Table 5—Daily Overall GHG emissions using newly introduced ProSolv HS8796 treatment

Current Case	Value	Input Unit	Emission Factor	Emission Unit	Factor	GHG emission (kgCO ₂ e)
Incumbent H ₂ S scavenger	0	kg	1.34	kgCO ₂ e/kg		0.00
Totes (IBC) 1000L	595	Unit	126.7	kgCO ₂ e/Unit		75,387
Triazine based H ₂ S Scavenger	0	kg	0.92	kgCO ₂ e/kg		0.00
Transport, freight, lorry, road trips	21600	metric ton*km	0.14	kgCO ₂ e/metric ton*km		2,951
PROSOLV HS8796	2499	kg	1.69	kgCO ₂ e/kg		4,215
Daily overall GHG Emissions						82,55

[Table 6](#) further quantifies the long-term environmental benefits over a 25-year field design lifetime, expressing the GHG reductions in terms of equivalent average passenger vehicles removed from roads.

This data demonstrates that ProSolv HS8796 not only improves operational efficiency but also significantly reduces the environmental footprint of the treatment process.

Table 6—Reductions of GHG values over time and estimated translation to the removal of average passenger vehicle of the road.

Duration of the Treatment	Reduction in Kg CO2 equivalent	Equivalent small SUV cars removed out of the road
Day	61,875	14
Month	1,856,258	404
Year	22,584,467	4910
5 years	112,922,337	24548
Asset estimated lifetime 25 years	564,611,683	122742

Additionally, improved pH stability and H₂S control have reduced iron sulfide (FeS) formation and other sedimentation problems, which often lead to frequent pigging, chemical flushing, and process downtime. By reducing these interventions, the system supports more sustainable operations and contributes to lower GHG emissions.

Conclusions

The field implementation of Veolia's ProSolv HS8796 has demonstrated exceptional performance in the treatment of highly sour streams, exhibiting approximately triple the scavenging capacity compared to the previous chemistry solution.

This superior performance has delivered multiple significant benefits across operational, safety, and environmental dimensions. The product has proven fully compatible with existing process streams, materials of construction, and injection points while maintaining robust asset integrity for sour production processing without requiring metallurgy upgrades or additional capital expenditure.

The solution ensures safe operations, protecting both personnel and environmental interests, while eliminating the need for off-spec gas flaring, thereby reducing environmental impact and carbon emissions.

Furthermore, ProSolv HS8796 has consistently achieved the critical performance target of maintaining H₂S levels below 5 ppm at the gas export line.

Improved H₂S control also contributed to reduced iron sulfide scale formation, minimizing associated risks such as flow assurance challenges, chemical overuse, and maintenance-related emissions.

This successful implementation aligns perfectly with both Daleel's community-focused sustainability initiatives and Veolia's strategic environmental objectives, representing a significant advancement in sustainable industrial chemistry applications.

Special Thanks

Special appreciation is extended to the dedicated team members at Daleel, whose invaluable support and collaboration were instrumental in the successful implementation of this project.

The success of this initiative would not have been possible without their meticulous attention to detail, thorough documentation, and proactive engagement throughout the entire process. Their commitment to safety, operational efficiency, and environmental stewardship exemplifies the highest standards of professional collaboration in the industry.

References

1. Speight, J.G., "The Chemistry and Technology of Petroleum," 6th Edition, CRC Press, ISBN 978-1138732705, pp. 235–276, 2019.
2. Craig, B.D., Anderson, D.S., "Handbook of Corrosion Data," 2nd Edition, ASM International, ISBN 978-0871705181, pp. 421–445, 1995.
3. Kelland, M.A., "Production Chemicals for the Oil and Gas Industry," 2nd Edition, CRC Press, ISBN 978-1439873793, pp. 325–358, 2014.
4. Kidnay, A.J., Parrish, W.R., McCartney, D.G., "Fundamentals of Natural Gas Processing," CRC Press, ISBN 978-1420085198, pp. 145–167, 2006.
5. NACE International, "Control of Internal Corrosion in Steel Pipelines and Piping Systems," NACE Standard SP0106-2018, Houston, TX, 2018.
6. API Technical Report 939-D, "Stress Corrosion Cracking of Carbon Steel in Corrosive Petroleum Refining Environments," Washington, DC, 2020.
7. ASTM International, "Standard Test Method for Measurement of Hydrogen Sulfide in the Vapor Phase Above Residual Fuel Oils," *ASTM D5705-12(2016)*, West Conshohocken, PA, 2016.
8. Abdel-Aal, H.K., Aggour, M.A., "Petroleum and Gas Field Processing," *Chemical Industries Series*, Marcel Dekker, ISBN 978-0824742843, pp. 187–212, 2003.
9. Environmental Protection Agency, "Greenhouse Gas Inventory Guidance: Direct Emissions from Mobile Combustion Sources," EPA-420-B-21-001, Washington, DC, 2021.
10. International Energy Agency, "Methane Emissions from Oil and Gas Operations," IEA Technical Report IEA-2021-01, Paris, France, 2021.